

Global AI Adoption in Education: A Comprehensive Global Performance Intelligence Report

A Data-Driven Analysis of Student, Teacher, and School-Level AI Adoption Across Global Education Systems (2015–2026)

Field	Detail
Student Name	Shubham Vitthal Patil
Program	Data Analytics with Tableau
Project Type	Individual
Tools	Tableau Desktop / Tableau Public
Phase	Project Initialization & Planning
Dataset	Global_AI_in_Education.csv (2015–2026, 10 countries, 6 regions, 1,360 records)
Date	03 July 2026

1. Define Problem Statement

Artificial intelligence is being adopted across classrooms, teaching staff, and school systems at very different speeds around the world. While student-facing tools such as ChatGPT, Google Gemini, Microsoft Copilot, and Khanmigo are increasingly common, education stakeholders still lack a clear, comparative view of how adoption differs across countries, regions, urban versus rural settings, and over time. Policymakers, school administrators, and edtech vendors need a structured way to see where AI adoption is accelerating, where it is lagging, and what factors — such as internet penetration, government policy, and curriculum integration — are driving the gap.

This project addresses the problem by analysing a global AI-in-education dataset spanning 2015 to 2026 using Tableau to uncover meaningful patterns and trends. The dataset includes key attributes such as Year, Month, Country, Region, Student AI Usage %, Teacher AI Usage %, School AI Adoption %, Average Daily AI Usage (minutes), Top AI Tool, Internet Penetration %, Education Index, Government AI Policy status, Urban/Rural AI Usage %, AI-in-Curriculum status, and Gender Gap in AI Usage %. By leveraging interactive visualizations, the project enables a deeper understanding of how these factors shape AI adoption outcomes across the global education landscape.

Through this analysis, the project identifies critical insights such as the fastest-adopting countries and regions, the equity gap between urban and rural AI usage, the relationship between government policy and curriculum integration, and the year-over-year acceleration in student and teacher AI usage. These findings help highlight both standout success stories and persistent gaps in global AI-education readiness.

Core Business Question

How can data-driven BI insights from global AI-in-education data empower policymakers, school leaders, and edtech stakeholders to make informed decisions about digital equity, curriculum policy, teacher enablement, and AI-tool investment?

Key Problem Dimensions

- AI usage distribution across students and teachers, by country, region, and year
- Identification of high-adoption countries/regions and leading AI tools in education
- Urban vs. rural AI usage gap and its relationship to internet penetration and education index
- Government AI policy status and its correlation with AI-in-curriculum adoption
- Trend analysis to detect year-over-year growth in AI usage and the gender gap in adoption

2. Project Planning

Work Breakdown Structure (Epics)

Epic / Task	Description
Data Collection & Extraction	Gather the global AI-in-education dataset covering Year, Month, Country, Region, Student/Teacher AI Usage, School AI Adoption, Daily Usage Minutes, Top AI Tool, Internet Penetration, Education Index, Government Policy, Urban/Rural Usage, Curriculum Status, and Gender Gap
Data Preparation	Clean, normalize, and structure the raw CSV data; handle missing Government AI Policy values, duplicates, and type inconsistencies for Tableau readiness
Data Visualizations	Build individual charts — adoption trend lines, regional heatmaps, country comparisons, urban/rural and gender-gap breakdowns, and top-tool usage charts
Dashboard	Assemble an interactive multi-view dashboard linking all visualizations with filters (year, country, region, policy status) and cross-highlighting
Story	Create a Tableau Story with annotated narrative slides guiding stakeholders through key AI-in-education adoption findings
Performance Testing	Validate dashboard load time, filter responsiveness, and cross-visualization accuracy
Web Integration	Publish to Tableau Public; integrate demo link and GitHub repository link into project documentation
Project Demonstration & Documentation	Record a walkthrough video, finalize README, and prepare presentation materials

Tools & Technology

- Tableau Desktop / Tableau Public — primary BI and visualization platform
- Microsoft Excel / CSV — dataset format for ingestion (Global_AI_in_Education.csv)
- GitHub — version control and project documentation repository

3. Proposed Solution

The solution leverages Tableau’s interactive BI capabilities to produce a data-driven platform that delivers targeted insights drawn from the global AI-in-education dataset. The analysis focuses on uncovering patterns in adoption trends, regional and country-level disparities, the urban-rural digital divide, and the policy-to-classroom pipeline.

Insight-Specific Solutions

- **Geographic & Regional Analysis** — country and region-wise comparison of student, teacher, and school AI adoption rates to identify leading and lagging markets.
- **Policy & Curriculum Insights** — analysis of government AI policy status (None / Draft / Active) against AI-in-curriculum adoption to reveal how policy translates into classroom practice.
- **Equity & Access Analysis** — urban vs. rural AI usage gap and gender gap in AI usage, cross-referenced with internet penetration and education index, to surface digital equity concerns.
- **Adoption & Tool Trend Analysis** — year-over-year growth in AI usage (2015–2026) and comparison of leading AI tools (ChatGPT, Google Gemini, Microsoft Copilot, Khanmigo) by region.

Project Deliverables

Deliverable	Description
Interactive Dashboard	Multi-tab Tableau dashboard with filters covering year, country/region, policy status, urban/rural, and AI tool dimensions
Tableau Story	Annotated story arc walking through key findings on adoption trends, policy impact, and equity gaps with data-backed recommendations
GitHub Repository	Full project source including dataset, workbook (.twbx), and documentation README
Demo Recording	Walkthrough video demonstrating dashboard interactions and key insights for mentor evaluation

Success Criteria

- Dashboard clearly addresses all key analytical dimensions with dedicated filter views
- Adoption trends, policy/curriculum relationships, urban-rural and gender-gap insights are accurate and reproducible from the source dataset

- Story narrative is coherent and communicates actionable policy and edtech recommendations
- Published Tableau Public workbook is accessible and loads within acceptable time
- GitHub repository is documented with README, dataset reference, and workbook file -e

Data Collection and Preprocessing Phase

Field	Detail
Student Name	Shubham Vitthal Patil
Date	03 July 2026
Team ID	Individual
Project Title	Global AI Adoption in Education: A Comprehensive Global Performance Intelligence Report
Maximum Marks	3 Marks

Data Quality Report Template

This Data Quality Report summarizes the quality assessment of the global AI-in-education dataset used in the project *Global AI Adoption in Education: A Comprehensive Global Performance Intelligence Report*. It documents identified data quality issues, their severity levels, and the resolution plans applied to ensure accurate, consistent, and reliable data for Tableau analysis and dashboard development.

Data Source	Data Quality Issue	Severity	Resolution Plan
Global_AI_in_Education.csv	15 of 16 columns fully populated with zero missing values (year, month, country, region, student_ai_usage_pct, teacher_ai_usage_pct, schools_ai_adoption_pct, avg_daily_ai_usage_min, top_ai_tool, internet_penetration_pct, education_index, urban_ai_usage_pct, rural_ai_usage_pct, ai_in_curriculum, gender_gap_ai_usage_pct)	Low	No action required. Verified using a null-count check across all columns — result: 0 nulls in 15 of 16 fields.
Global_AI_in_Education.csv	government_ai_policy	Moderate	Replace nulls with an

Data Source	Data Quality Issue	Severity	Resolution Plan
v	contains 441 missing values (32.4% of 1,360 records). These represent country/period combinations with no formal AI-in-education policy rather than a data-collection error.	High	explicit “None” category using IFNULL()/COALESCE so all three policy states — None, Draft, Active — are represented consistently in filters, tooltips, and color legends.
Global_AI_in_Education.csv	No duplicate records found. All 1,360 country-year-month combinations are unique with zero duplication across the entire dataset.	Low	No action required. Duplicate check on the country-year-month key returned 0 duplicates. Dataset integrity confirmed for all records.
Global_AI_in_Education.csv	Region-level totals are unevenly weighted: Asia contains 408 records (3 countries — China, India, Pakistan) while Africa, South America, and Oceania each contain 136 records (1 country each). A simple AVG() by region can misrepresent regional performance.	Moderate	Add a per-country average measure alongside the region total, and label region-level charts clearly as “country-weighted” where relevant. Consider a parameter to toggle between SUM and per-country AVG views.
Global_AI_in_Education.csv	education_index is stored as float64 but ranges narrowly between 0.52 and 0.94, which can visually flatten differences between countries on a default 0–1 axis.	Low	Set a custom axis range (e.g. 0.4–1.0) or add reference lines/bands in Tableau to preserve visible differentiation between countries.
Global_AI_in_Education.csv	year and month are stored as separate int64 columns. A month value alone (e.g. “6”) is ambiguous across the 2015–2026 span and may mislead trend charts if	Low	Create a composite calculated field Period = STR([year]) + “-” + RIGHT(“0” + STR([month]), 2), formatted as YYYY-MM, for use on all

Data Source	Data Quality Issue	Severity	Resolution Plan
	plotted independently of year.		time-series axes.
Global_AI_in_Education.csv	ai_in_curriculum is stored as “Yes”/“No” text rather than a boolean flag, which reduces readability for rate calculations and non-technical dashboard viewers.	Low	Create a Tableau calculated field: IF [ai_in_curriculum] = “Yes” THEN 1 ELSE 0 END, and use it to compute a Curriculum Integration Rate for KPI cards and trend lines.
Global_AI_in_Education.csv	top_ai_tool records only a single leading tool per country-month (ChatGPT, Google Gemini, Microsoft Copilot, or Khanmigo), which limits analysis of tool co-adoption or secondary-tool usage.	Low	Document this as a stated assumption in the project README: the field represents the leading tool only, not the full tool stack in use. No further correction possible without additional source data.
Global_AI_in_Education.csv	No standardized composite keys exist for cross-dimensional analysis (e.g. no single column combining country + region, or government_ai_policy + ai_in_curriculum), limiting direct multi-attribute segmentation.	Low	Create derived calculated columns in Tableau: Location Label = [country] + “ - ” + [region]; Policy-Curriculum Alignment = [government_ai_policy (clean)] + “ - ” + [ai_in_curriculum]. Use these for heatmaps and cross-segment filters.
Global_AI_in_Education.csv	urban_ai_usage_pct and rural_ai_usage_pct are independent columns (ranges 0–70 and 0–59 respectively); the derived Urban-Rural Usage Gap could theoretically be negative, which would need review rather than	Low	Add a calculated field Urban-Rural Usage Gap = [urban_ai_usage_pct] – [rural_ai_usage_pct], validate the sign distribution, and flag any negative values for manual review

Data Source	Data Quality Issue	Severity	Resolution Plan
	being treated as an error by default.		rather than automatic exclusion.

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Data Collection and Preprocessing Phase

Field	Detail
Student Name	Shubham Vitthal Patil
Date	03 July 2026
Team ID	Individual
Project Title	Global AI Adoption in Education: A Comprehensive Global Performance Intelligence Report
Maximum Marks	2 Marks

Data Collection Plan & Raw Data Sources Identification Template

This report outlines the data collection plan and raw data sources identified for the project *Global AI Adoption in Education: A Comprehensive Global Performance Intelligence Report*. It ensures meticulous data curation and integrity by documenting the source, format, size, and access details of the dataset used for analysis and Tableau dashboard development.

Data Collection Plan Template

Section	Description
Project Overview	This project analyses global AI adoption trends in education using a dataset of 1,360 monthly records across 16 attributes, including student, teacher, and school-level AI usage, government policy status, curriculum integration, and urban/rural usage split. The objective is to build an interactive Tableau dashboard that uncovers adoption trends, policy impact, tool preferences, and digital-equity gaps to support data-driven decisions for policymakers, school leaders, and edtech stakeholders.
Data Collection Plan	The primary dataset was sourced from Kaggle — a publicly available dataset titled “Global AI in Education Dataset (2015–2026)” (kaggle.com/datasets/abidhussai512/global-ai-in-education-dataset-20152026) in CSV format containing 1,360 rows and 16 columns. No additional external APIs or web scraping were required, as the dataset comprehensively covers all required dimensions: adoption attributes (student_ai_usage_pct, teacher_ai_usage_pct, schools_ai_adoption_pct), infrastructure attributes (internet_penetration_pct, education_index), policy and curriculum attributes (government_ai_policy, ai_in_curriculum), equity

Section	Description
	attributes (urban_ai_usage_pct, rural_ai_usage_pct, gender_gap_ai_usage_pct), and contextual attributes (country, region, top_ai_tool, year, month). The dataset was downloaded directly and stored locally for preprocessing and Tableau ingestion.
Raw Data Sources Identified	One primary raw data source was identified: Global_AI_in_Education.csv — a structured CSV file containing 1,360 monthly country-level records with 16 columns covering AI adoption, infrastructure, policy, and equity metrics across 10 countries and 6 regions from 2015 to 2026. The dataset is largely clean and publicly accessible with no licensing restrictions for academic and analytical use, with one column (government_ai_policy) requiring null-handling during preprocessing.

Raw Data Sources Template

Source Name	Description	Location/URL	Format	Size	Access Permissions
Global_AI_in_Education.csv	1,360 monthly records with 16 columns: year, month, country, region, student_ai_usage_pct, teacher_ai_usage_pct, schools_ai_adoption_pct, avg_daily_ai_usage_min, top_ai_tool, internet_penetration_pct, education_index, government_ai_policy, urban_ai_usage_pct, rural_ai_usage_pct, ai_in_curriculum, gender_gap_ai_usage_pct.	kaggle.com/datasets/abidhussai512/global-ai-in-education-dataset-20152026	CSV	0.10 MB (1,360 rows x 16 columns)	Public

Source Name	Description	Location/URL	For mat	Size	Access Permis sions
	Covers 10 countries across 6 regions, spanning 2015–2026. Student AI usage ranges 0–65% (avg 30.99%). 15 of 16 columns fully populated; government_ai_policy has 441 nulls (32.4%), representing “no formal policy” periods.				
N/A	No additional datasets used for this project.	N/A	N/A	N/A	N/A
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Data Collection and Preprocessing Phase

Field	Detail
Student Name	Shubham Vitthal Patil
Date	03 July 2026
Team ID	Individual
Project Title	Global AI Adoption in Education: A Comprehensive Global Performance Intelligence Report
Maximum Marks	10 Marks

Data Exploration and Preprocessing Template

Identifies data sources, assesses quality issues such as missing values and category imbalance, and implements resolution plans to ensure accurate and reliable analysis of

global AI adoption trends, policy impact, and equity gaps in education across 1,360 records and 16 attributes.

Section	Description
Data Overview	The dataset contains 1,360 global AI-in-education records with 16 columns: year, month, country, region, student_ai_usage_pct, teacher_ai_usage_pct, schools_ai_adoption_pct, avg_daily_ai_usage_min, top_ai_tool, internet_penetration_pct, education_index, government_ai_policy, urban_ai_usage_pct, rural_ai_usage_pct, ai_in_curriculum, and gender_gap_ai_usage_pct. It covers 10 countries across 6 regions (Asia, North America, Europe, South America, Africa, Oceania). Records span monthly observations from 2015 to 2026. Student AI usage ranges from 0% to 65% (avg 30.99%), teacher AI usage from 0% to 60% (avg 29.91%), and school-level AI adoption from 0% to 62% (avg 29.41%). 15 of 16 columns are fully populated; government_ai_policy contains 441 missing entries (32.4%).
Data Cleaning	No duplicate records found — all 1,360 country-year-month combinations are unique. Missing values are confined to government_ai_policy (441 of 1,360 rows, 32.4%), which represent countries/periods with no formal AI-in-education policy rather than a data-collection error; these are resolved by mapping nulls to an explicit “None” category alongside the existing “Draft” and “Active” statuses. Categorical columns (country, region, top_ai_tool, ai_in_curriculum) contain clean, consistent labels with no mixed-case or spelling variants. Numerical columns (student/teacher/school AI usage %, internet penetration %, urban/rural usage %, gender gap %, education_index) fall within expected bounds. Dataset is confirmed clean and Tableau-ready after null handling.
Data Transformation	Filter records by country, region, year, and government_ai_policy for focused segment analysis. Sort by student_ai_usage_pct descending to identify leading adoption markets, and by schools_ai_adoption_pct to surface leading school systems. Pivot data to create country-wise and region-wise year-over-year aggregations of average AI usage and adoption. Create calculated columns: Urban-Rural Usage Gap = (urban_ai_usage_pct – rural_ai_usage_pct); Student-Teacher Adoption Ratio = (student_ai_usage_pct / teacher_ai_usage_pct); Adoption Tier (Low: 0–25, Mid: 26–45, High: 46–65) based on student_ai_usage_pct; Policy Status (Clean) = IFNULL([government_ai_policy], “None”).
Data Type Conversion	year and month confirmed as int64 and used to build time-series axes. student_ai_usage_pct, teacher_ai_usage_pct, schools_ai_adoption_pct, avg_daily_ai_usage_min, internet_penetration_pct, urban_ai_usage_pct, rural_ai_usage_pct, and gender_gap_ai_usage_pct (int64, whole-number percentages/minutes) are correctly typed for aggregation. education_index (float64, range 0.52–0.94) confirmed for precise

Section	Description
	averaging and correlation analysis. country, region, top_ai_tool, government_ai_policy, and ai_in_curriculum confirmed as str and loaded as Tableau Dimensions. No type mismatches detected.
Column Splitting and Merging	Merge year and month into a single Period field (formatted YYYY-MM) to support clean time-series axes across the 2015–2026 span. Merge country and region into a Location Label (e.g. “India – Asia”) to support geographic drill-downs. Convert ai_in_curriculum (Yes/No) into a Boolean Curriculum Integrated flag for rate calculations. Combine government_ai_policy with ai_in_curriculum into a Policy-Curriculum Alignment tag (e.g. “Active – Yes”) to explore how policy adoption translates into classroom practice.
Data Modeling	Primary table: Education AI Adoption (1,360 rows, 16 columns) used as single source of truth. Tableau calculated measures: Avg Student Usage = $\text{AVG}(\text{student_ai_usage_pct}) = 30.99\%$; Avg Teacher Usage = 29.91% ; Avg School Adoption = 29.41% ; Avg Internet Penetration = 74.70% ; Avg Education Index = 0.776 ; Avg Gender Gap = 5.41 points; Active Policy Rate $\approx 34.85\%$ ($474/1,360$); Draft Policy Rate $\approx 32.72\%$ ($445/1,360$); No-Policy Rate $\approx 32.43\%$ ($441/1,360$); Curriculum Integration Rate $\approx 50.51\%$ ($687/1,360$). Country distribution is evenly balanced at 136 records each across all 10 countries. Region distribution is skewed toward Asia (408 records, 30%), which spans three countries (China, India, Pakistan) versus one to two countries for other regions.
Save Processed Data	Original dataset (Global_AI_in_Education.csv, 1,360 rows x 16 columns) saved as primary source — no rows removed, as the dataset required only null-handling on one column. Processed CSV with added calculated columns (Urban-Rural Usage Gap, Student-Teacher Adoption Ratio, Adoption Tier, Period, Location Label, Policy-Curriculum Alignment, Policy Status Clean) exported as ai_education_processed.csv for Tableau ingestion. Tableau workbook saved as global_ai_education_analysis.twbx with embedded data source. All files version-controlled in the GitHub repository with preprocessing steps documented in README.md.

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GLOBAL AI ADOPTION IN EDUCATION

Decoding Global AI Adoption Across Education Systems

A Tableau-Based Business Intelligence Deep Dive into Global AI Adoption in Education (2015–2026)

Field	Detail
Project Title	Global AI Adoption in Education: A Comprehensive Global Performance Intelligence Report
Prepared By	Shubham Vitthal Patil
Program	SkillWallet Data Analytics Program
Tool Used	Tableau (Desktop / Public)
Dataset	Global_AI_in_Education.csv (Kaggle: Global AI in Education Dataset 2015–2026)
Tableau Public Dashboard	View Live Dashboard
Tableau Public Story	View Live Story
GitHub Repository	github.com/shubham-patil/global-ai-adoption-in-education <i>(replace with your actual repository URL once created)</i>
Document Type	End-to-End Project Documentation
Date	3 July 2026

1. Executive Summary

Global AI Adoption in Education is an end-to-end business intelligence project built in Tableau to analyze how artificial intelligence has spread through classrooms, teaching staff, and school systems worldwide between 2015 and 2026. The project ingests a curated global dataset — spanning student, teacher, and school-level AI usage, government policy status, curriculum integration, and urban/rural equity splits — and transforms it into a set of interactive dashboards and a guided data story.

The final deliverable consists of 8 supporting worksheets, 1 consolidated interactive dashboard, and a 5-page narrative story, all cross-filtered by Country, Region, Year, and Government AI Policy. The analysis is designed to serve three distinct audiences: education policymakers deciding where to focus AI-policy investment, school administrators and teachers benchmarking their own adoption against global peers, and edtech analysts tracking macro adoption and tool-preference trends.

Headline findings include: global student AI usage has grown roughly twelvefold, from about 5% in 2015 to over 60% in 2026; adoption growth is essentially uniform across all ten countries in the dataset, clustering tightly between 30.6% and 31.6% on average, regardless of internet penetration or education-index level; a formally “Active” government AI policy shows almost no measurable relationship with whether AI actually appears in the curriculum; and the urban-rural usage gap widened sharply in the dataset’s early years and has since plateaued around 12 percentage points rather than closing.

2. Problem Statement

2.1 Background

Artificial intelligence has moved from a novelty in a handful of classrooms to a routine part of student and teacher workflows in just over a decade. Adoption data is fragmented across national education ministries, edtech vendor reports, and academic surveys, making it difficult for any single stakeholder to form an accurate, comparative, data-backed picture of how AI adoption is really progressing across countries, regions, and demographic groups.

2.2 Core Problem

There is no single, interactive, filterable view that lets a user answer practical questions such as:

- Which countries and regions are leading — or lagging — in student, teacher, and school-level AI adoption?
- Does a country’s internet penetration or education index actually predict how quickly it adopts AI in education?
- Does having a formal government AI policy translate into AI actually appearing in the curriculum?
- Is the urban-rural AI usage gap closing over time, or persisting?
- Which AI tools (ChatGPT, Google Gemini, Microsoft Copilot, Khanmigo) are gaining or losing ground by region?

2.3 Objective of This Project

To design and build a Tableau dashboard and story that consolidates global AI-in-education data into clear, interactive, and filterable visualizations — enabling data-driven decisions for policymakers, school leaders, and edtech analysts without requiring any manual spreadsheet analysis.

2.4 Target Stakeholders

Stakeholder	Key Question Answered
Education Policymaker	Is our national AI policy actually translating into classroom adoption, and how do we compare globally?
School Administrator / Teacher	How does our school’s likely AI adoption level compare to global and regional benchmarks?
EdTech Analyst / Vendor	Where is AI-in-education demand concentrated, which tools are winning, and how is that shifting year over year?

3. Project Objectives

- Consolidate the raw global AI-in-education data into a single, clean Tableau data source.
- Engineer calculated fields that convert raw percentages and text flags into decision-ready metrics (e.g., adoption tiers, usage gaps, policy-curriculum alignment).

- Build a set of focused worksheets, each answering one specific business question.
- Combine these worksheets into one interactive, filterable dashboard.
- Construct a 5-page guided Story that walks a non-technical viewer through the same insights in a logical narrative order.
- Extract and document clear, quantified insights that are useful to each of the three target stakeholders.

4. Dataset Overview

The primary data source is a single flat table, `Global_AI_in_Education`, connected directly inside Tableau (visible in the Data pane as “`Global_AI_in_Educat...`”).

4.1 Dataset Scale

- Total monthly country-level records analyzed: 1,360
- Coverage: 10 countries across 6 regions, monthly observations from January 2015 through April 2026
- Countries: United States, United Kingdom, China, India, Germany, Brazil, Pakistan, Nigeria, Canada, Australia (136 records each)
- Regions: Asia (408 records, 3 countries), North America (272), Europe (272), South America (136), Africa (136), Oceania (136)
- AI tools tracked: 4 (ChatGPT, Google Gemini, Microsoft Copilot, Khanmigo)
- Government policy statuses: 3 (Active, Draft, and unrecorded/“None”)

4.2 Raw Fields (as found in the Data pane)

Field Name	Description
Year / Month	Temporal fields used for trend analysis (2015–2026)
Country / Region	Geographic dimensions used for country- and region-level comparison
Student Ai Usage Pct	Percentage of students actively using AI tools for learning
Teacher Ai Usage Pct	Percentage of teachers actively using AI tools for teaching/admin tasks
Schools Ai Adoption Pct	Percentage of schools with institutional AI adoption
Avg Daily Ai Usage Min	Average daily minutes of AI tool usage per user
Top Ai Tool	The leading AI tool in use for that country-month (ChatGPT, Google Gemini, Microsoft Copilot, Khanmigo)
Internet Penetration Pct	Percentage of the population with internet access
Education Index	A normalized 0–1 score representing overall education-system quality
Government Ai Policy	Policy status: Active, Draft, or blank (no formal policy)
Urban Ai Usage Pct / Rural Ai Usage Pct	AI usage split by urban and rural populations
Ai In Curriculum	Yes/No flag indicating whether AI is formally part of the

Field Name	Description
	curriculum
Gender Gap Ai Usage Pct	Percentage-point gap in AI usage between genders

5. Data Preparation & Preprocessing

Before building any visualization, the raw dataset was reviewed and enriched with calculated fields inside Tableau so that every downstream worksheet could reuse consistent, decision-ready metrics rather than raw numbers.

5.1 Data Cleaning Steps

- Verified there were no duplicate country-year-month combinations that would inflate counts (0 duplicates across 1,360 records).
- Standardized categorical text fields (Country, Region, Top AI Tool, Government AI Policy, AI In Curriculum) to consistent casing/spelling so Tableau does not split a single real-world category into multiple bars.
- Confirmed numeric fields (Student/Teacher/School AI Usage %, Internet Penetration %, Education Index) were correctly typed as numbers, not strings, so aggregations (SUM/AVG) compute correctly.
- Resolved 441 blank Government AI Policy values (32.4% of records) by mapping them to an explicit “None” category rather than leaving them as nulls, so all three policy states are represented consistently in filters and legends.

5.2 Calculated Fields Created

Calculated Field	Purpose
Policy Status (Clean)	IFNULL([Government Ai Policy], “None”) — ensures every record has an explicit policy label for filtering and color legends
Urban-Rural Usage Gap	Urban Ai Usage Pct – Rural Ai Usage Pct — quantifies the digital-equity divide by country and year
Adoption Tier	Buckets Student Ai Usage Pct into Low (0–25), Mid (26–45), and High (46–65) for simplified comparison
Curriculum Integrated (Flag)	IF [Ai In Curriculum] = “Yes” THEN 1 ELSE 0 END — enables Curriculum Integration Rate calculations
Policy-Curriculum Alignment	[Policy Status (Clean)] + “-” + [Ai In Curriculum] — flags whether policy adoption is matched by classroom integration
Location Label	[Country] + “-” + [Region] — supports geographic drill-downs and map-style visuals
Period	STR([Year]) + “-” + RIGHT(“0”+STR([Month]),2) — a clean YYYY-MM field for all time-series axes

5.3 Adoption Tier Binning

Raw Student Ai Usage Pct values were grouped into an Adoption Tier dimension (Low / Mid / High) so that comparisons across countries and years could be made at a band level

rather than requiring a continuous axis — this makes cross-category comparison in bar charts far more readable.

5.4 Filtering Strategy

Four filters were applied globally at the data-source level so that every worksheet (and therefore the dashboard) stays in sync:

- Country — allows drill-down into a specific country or a global (All) view.
- Region — allows filtering by continent-level grouping.
- Year — allows the viewer to isolate a specific year or range within 2015–2026.
- Government AI Policy (clean) — allows the viewer to focus on Active, Draft, or None policy states.

In addition, a Top-N context filter was layered on top of individual views to keep charts focused on the most significant categories:

- Top 5 by AVG([Student Ai Usage Pct]) — restricts country-comparison views to the five leading adopters.

6. Tools & Methodology

6.1 Tools Used

- Tableau Public / Tableau Desktop — for data connection, calculated fields, visualization, dashboarding, and storytelling.
- GitHub — for version control and hosting the project repository, including the dataset, workbook, documentation, and screenshots.

6.2 Repository Structure

Folder	Contents
data/	Raw and cleaned dataset(s) used for the analysis
workbook/	The Tableau (.twbx) workbook file
docs/	Project documentation, including this document
assets/screenshots/	Exported screenshots of every worksheet, the dashboard, and the story

6.3 Workbook Structure

The Tableau workbook is organized into 8 worksheets, one combined Dashboard (“Dashboard 1”), and one Story (“Story 1”):

Sheet	Chart Title	Chart Type
Sheet 1	AI Adoption Trend by Year (Student / Teacher / School)	Multi-line time series, 2015–2026
Sheet 2	Country vs Student AI Usage (%)	Horizontal bar chart, 10 countries

Sheet	Chart Title	Chart Type
Sheet 3	Urban vs Rural AI Usage Gap by Country	Diverging / dumbbell bar chart
Sheet 4	Key Performance Indicators (KPIs)	Text/KPI summary panel
Sheet 5	Government Policy vs Curriculum Integration	Stacked bar chart, 3 policy states
Sheet 6	Top AI Tool Adoption by Region	Stacked bar / treemap, 6 regions x 4 tools
Sheet 7	Internet Penetration & Education Index vs AI Adoption	Scatter plot with trend line
Sheet 8	Gender Gap in AI Usage by Year	Line chart, 2015–2026

7. Dashboard Deep Dive

Five of the eight worksheets are combined into a single interactive dashboard, titled “Global AI Adoption in Education: Decoding the Adoption Curve”, with a shared global filter panel (Country, Region, Year, Government AI Policy) positioned on the right-hand side.

7.1 Tile 1 — AI Adoption Trend by Year

Chart type: Multi-line chart. Columns = Year; Rows = AVG(Student/Teacher/School AI Usage %).

Purpose: To show how quickly AI adoption has grown across the three adoption layers (students, teachers, schools) since 2015.

Observation: All three lines rise together in near lock-step — from roughly 4–5% in 2015 to 59–61% in 2026 — with student usage consistently leading teacher and school-level adoption by a small, stable margin throughout the period.

7.2 Tile 2 — Country vs Student AI Usage (%)

Chart type: Horizontal bar chart. Rows = Country; Columns = AVG(Student AI Usage Pct).

Purpose: To test whether some countries are meaningfully outpacing others in AI-in-education adoption.

Observation: Bar lengths across all ten countries are visually close, ranging only from 30.60% (Brazil) to 31.61% (United States) — a spread of about one percentage point — indicating adoption is a genuinely global, near-uniform phenomenon rather than being led by a handful of front-runner nations.

7.3 Tile 3 — Key Performance Indicators (KPIs)

Chart type: Text table summarizing four headline metrics using Measure Names / Measure Values.

Values shown: 30.99% average student AI usage; 74.70% average internet penetration; 0.776 average education index; 50.51% curriculum integration rate (687 of 1,360 records).

Purpose: To give any viewer an at-a-glance adoption and readiness snapshot before drilling into details.

7.4 Tile 4 — Urban vs Rural AI Usage Gap by Country

Chart type: Dumbbell / diverging bar chart. Rows = Country; two marks per row for Urban and Rural AI Usage %.

Observation: Every country shows a gap of roughly 11.6 to 12.2 percentage points between urban and rural AI usage, with Pakistan (12.24 pts) and Canada (12.22 pts) showing the widest gaps and India (11.57 pts) the narrowest — the gap is remarkably consistent regardless of a country's overall internet penetration.

7.5 Tile 5 — Government Policy vs Curriculum Integration

Chart type: Stacked bar chart. Columns = Policy Status (Active / Draft / None); Rows = % of records with AI In Curriculum = Yes.

Observation: Curriculum integration sits at 52.3% under an Active policy, 51.9% with no formal policy at all, and only 47.2% under a Draft policy — a strikingly flat relationship that suggests a formally “Active” government AI policy is not, on its own, a strong predictor of whether AI actually reaches the classroom.

7.6 Dashboard Interactivity

The dashboard supports live cross-filtering: adjusting the Country, Region, Year, or Government AI Policy selector in the right-hand filter panel instantly updates all five tiles simultaneously, allowing a viewer to, for example, isolate the picture for a single country's adoption curve across a chosen policy state.

8. Story Walkthrough — “Decoding the Adoption Curve (2015–2026)”

Beyond the single-screen dashboard, a 5-page Tableau Story was built to walk a viewer through the same data in a logical narrative sequence — starting broad (the global growth curve) and progressively narrowing to a specific, surprising conclusion (policy versus practice, and persistent equity gaps).

8.1 Story Page 1 — AI Adoption Trend by Year

A full time-series view of student, teacher, and school AI adoption from 2015 to 2026. Student usage climbs from 4.98% in 2015 to 60.75% by 2026 — roughly a twelvefold

increase — with teacher (59.00%) and school-level (59.25%) adoption converging closely behind by the final year.

8.2 Story Page 2 — Country vs Student AI Usage (%)

A 10-bar chart ranking every country by average student AI usage, led by the United States (31.61%) and Canada (31.07%), tapering down to Brazil (30.60%) at the lower end — a visibly tight cluster rather than a wide spread.

8.3 Story Page 3 — Internet Penetration & Education Index vs AI Adoption

Reiterates the dashboard's scatter view within the story's narrative flow. Internet Penetration and Education Index correlate strongly with each other ($r \approx 0.98$, as expected), but neither correlates meaningfully with Student, Teacher, or School AI Usage ($r \approx 0.00$ in all cases) — undermining the assumption that better-connected or higher-scoring education systems necessarily adopt AI faster.

8.4 Story Page 4 — Government Policy vs Curriculum Integration

Repeats the Policy vs Curriculum view, driving home the insight that curriculum integration rates are nearly identical whether a country has an Active policy (52.3%), a Draft policy (47.2%), or no recorded policy at all (51.9%) — a genuinely counter-intuitive finding worth flagging to policymakers.

8.5 Story Page 5 — Equity Gaps Over Time (Closing Insight)

The story's final and most important page: the urban-rural usage gap widened from 7.68 points in 2015 to roughly 12–13 points by 2018, and has stayed essentially flat there ever since (12.98 points in 2026); the gender gap in AI usage has similarly held steady between 5.05 and 5.91 points across the entire period, with no clear narrowing trend.

This is the story's key "punchline": overall AI adoption has grown enormously in twelve years, but the equity gaps that existed early on — urban vs. rural, and gender — have not closed alongside it. Growth in AI adoption has been additive rather than equalizing, implying that policy and investment aimed specifically at rural and gender-gap closure, not just overall AI rollout, will be needed to bring these gaps down.

9. Key Insights & Findings

- Global AI adoption in education has grown roughly twelvefold in twelve years: student usage rose from 4.98% (2015) to 60.75% (2026), with teacher and school-level adoption following an almost identical curve.
- Country-level adoption is remarkably uniform: all ten countries cluster between 30.6% and 31.6% average student usage — a spread of about one percentage point — meaning no single country is a clear runaway leader or laggard.
- Internet penetration and education index do not predict adoption speed: despite correlating strongly with each other ($r \approx 0.98$), neither shows any meaningful correlation with student, teacher, or school AI usage ($r \approx 0.00$) — adoption appears

to be driven primarily by time/global trend rather than digital infrastructure or education quality.

- A formally “Active” government AI policy is a weak predictor of curriculum integration: curriculum-integration rates are nearly flat across Active (52.3%), Draft (47.2%), and None (51.9%) policy states — a genuinely counter-intuitive finding worth flagging to policymakers and analysts.
- Equity gaps persist despite overall growth: the urban-rural usage gap widened from 7.68 points (2015) to about 12–13 points by 2018 and has stayed there since; the gender gap has held flat between roughly 5 and 6 points throughout — neither gap is closing as adoption rises.
- AI tool preference varies modestly by region: Asia and Africa show the broadest four-way split among ChatGPT, Google Gemini, Microsoft Copilot, and Khanmigo, while the most recent months of 2026 (a smaller four-month sample) show ChatGPT pulling ahead — a trend worth monitoring rather than treating as fully confirmed given the limited sample.

9.1 Stakeholder-Specific Takeaways

Stakeholder	Actionable Takeaway
Education Policymaker	Don’t assume an “Active” policy alone will move the curriculum-integration needle; pair policy rollout with direct classroom implementation support and monitoring.
School Administrator / Teacher	Your school’s adoption trajectory likely mirrors the global curve regardless of local internet infrastructure — benchmark against the ~31% country-average cluster, not just well-connected peers.
EdTech Analyst / Vendor	Target rural and gender-gap-focused product design specifically — general AI rollout has not closed these gaps on its own; watch regional tool-preference shifts, especially any emerging ChatGPT consolidation.

10. Challenges Faced & Learnings

10.1 Data-Level Challenges

- Handling a heavily null policy field: Government AI Policy was blank for 32.4% of records; mapping these to an explicit “None” category (rather than dropping rows or leaving nulls) was essential so filters and legends stayed usable.
- Uneven region-to-country mapping: Asia contains three countries (China, India, Pakistan) versus one to two for other regions, so region-level totals needed a per-country average alongside straight sums to avoid misrepresenting regional performance.
- Distinguishing correlation from trend: because adoption metrics, urban/rural usage, and year all move together ($r \approx 0.97\text{--}0.99$), it took a dedicated scatter view against Internet Penetration and Education Index specifically to confirm that infrastructure/quality — not just time — has essentially no independent relationship with adoption.

10.2 Design Challenges

- Choosing a chart type that could show the urban-rural gap as a gap, not just two overlapping bars, led to a dumbbell/diverging-bar design instead of a simple grouped bar chart.
- Deciding between a Story (linear narrative) and a Dashboard (all-at-once view) — ultimately both were built, since they serve different audiences: the Dashboard for self-service exploration, the Story for guided presentation.

10.3 Key Learnings

- Calculated fields (Policy Status Clean, Urban-Rural Usage Gap, Policy-Curriculum Alignment) are what turn a raw spreadsheet into a decision-support tool — most of the analytical value is created before a single chart is drawn.
- A flat policy-vs-curriculum relationship, and a persistent (not closing) equity gap, are genuinely newsworthy insights — they emerged only after deliberately building dedicated charts for them, reinforcing the value of testing assumptions explicitly rather than only reporting the obvious growth-trend metric.

11. Conclusion & Future Scope

Global AI Adoption in Education successfully consolidates a fragmented, fast-moving global adoption picture into a single, interactive Tableau experience that serves policymakers, school leaders, and edtech analysts alike. The project demonstrates the full BI workflow — from raw data and calculated-field engineering, through worksheet design, to dashboard assembly and narrative storytelling — culminating in clear, quantified, and in some cases counter-intuitive insights (e.g., the near-zero link between infrastructure/education quality and adoption speed, and the flat policy-to-curriculum relationship).

Future scope includes extending the dataset beyond 2026 as new months become available, incorporating additional equity dimensions (e.g., income level or school funding tier) if such data becomes available, and building a predictive view that forecasts when the urban-rural and gender gaps might realistically close under different policy-investment scenarios. -e

Business Question and Visualization Report

Global AI Adoption in Education

Field	Detail
Student Name	Shubham Vitthal Patil
Project Name	Global AI Adoption in Education: A Comprehensive Global Performance Intelligence Report
Team ID	Individual

Field	Detail
Tools	Tableau Desktop / Tableau Public

Visualization development transforms raw data into graphical formats that facilitate understanding, analysis, and decision-making. For this project, Tableau was used to convert **1,360 monthly records (2015–2026) across 10 countries, 6 regions, and 16 attributes** into 8 focused worksheets, 1 consolidated dashboard, and a 5-page story. The visualizations enable stakeholders — educators, policymakers, and researchers — to gain insight into global AI adoption trends, equity gaps, and the relationship between policy and actual classroom usage.

Business Questions and Visualisation

1. What are the overall key performance metrics of global AI adoption in education?

- **Visualization:** Text Table (KPI Summary) showing Avg. Student AI Usage: 30.99% | Avg. Internet Penetration: 74.70% | Avg. Education Index: 0.776 | Curriculum Integration Rate: 50.51% (687 of 1,360 records).
- **Purpose:** To provide an overall snapshot of the global dataset before users explore detailed visualizations.

2. How has AI adoption grown across students, teachers, and schools from 2015 to 2026?

- **Visualization:** Multi-line time series showing `student_ai_usage_pct`, `teacher_ai_usage_pct`, and `schools_ai_adoption_pct` over time.
- **Observation:** All three adoption layers rise together in near lock-step — from roughly 4.98% (student) in 2015 to 60.75% by 2026, a roughly twelvefold increase — with teacher (59.00%) and school-level (59.25%) adoption converging closely behind by the final year.
- **Purpose:** To visualize the trajectory of AI adoption and confirm whether growth is consistent across the three stakeholder layers.

3. Are some countries meaningfully outpacing others in AI adoption?

- **Visualization:** Horizontal bar chart ranking countries by average `student_ai_usage_pct`.
- **Observation:** Adoption is remarkably uniform globally — bar lengths range only from 30.60% (Brazil) to 31.61% (United States), a spread of about one percentage point. No country is a clear runaway leader or laggard.
- **Purpose:** To test the assumption that certain countries dominate AI-in-education adoption — and disprove it with the data.

4. Does internet penetration or education quality predict how fast a country adopts AI?

- **Visualization:** Scatter plot with trend line comparing `internet_penetration_pct` and `education_index` against `student_ai_usage_pct`.
- **Observation:** Internet penetration and education index correlate strongly with each other ($r \approx 0.98$, as expected), but neither correlates meaningfully with student,

teacher, or school AI usage ($r \approx 0.00$ in all cases) — adoption is being driven primarily by time/global trend, not digital infrastructure or education-system quality.

- **Purpose:** To directly challenge the assumption that better-connected or higher-scoring education systems adopt AI faster.

5. How large is the urban-rural AI usage gap, and is it closing over time?

- **Visualization:** Dumbbell / diverging bar chart showing Urban vs. Rural AI usage % by country, alongside a year-over-year gap trend line.
- **Observation:** The gap widened sharply from 7.68 points in 2015 to roughly 12–13 points by 2018, and has stayed essentially flat since (12.98 points in 2026). By country, Pakistan (12.24 pts) and Canada (12.22 pts) show the widest gaps; India (11.57 pts) the narrowest.
- **Purpose:** To test whether overall adoption growth is closing the urban-rural equity gap — it is not.

6. Does a formal government AI policy translate into curriculum integration?

- **Visualization:** Stacked bar chart comparing government_ai_policy status (Active / Draft / None) against the % of records with ai_in_curriculum = Yes.
- **Observation:** Curriculum integration sits at 52.3% under an Active policy, 51.9% with no formal policy at all, and only 47.2% under a Draft policy — a strikingly flat relationship, showing a formally “Active” policy is a weak predictor of real classroom integration.
- **Purpose:** To evaluate whether policy is driving or merely coinciding with classroom adoption.

7. Which AI tool dominates in each region?

- **Visualization:** Stacked bar chart / treemap showing top_ai_tool share by region.
- **Observation:** Asia and Africa show the broadest four-way split among ChatGPT, Google Gemini, Microsoft Copilot, and Khanmigo. The most recent months of 2026 (a smaller four-month sample) show ChatGPT pulling ahead — a trend worth monitoring rather than treating as fully confirmed given the limited sample size.
- **Purpose:** To understand the competitive landscape of AI tools across different educational markets.

8. Is there a gender gap in AI usage, and is it closing?

- **Visualization:** Line chart showing gender_gap_ai_usage_pct by year, 2015–2026.
- **Observation:** The gender gap has held steady between 5.05 and 5.91 points across the entire period, with no clear narrowing trend — mirroring the persistence seen in the urban-rural gap.
- **Purpose:** To monitor equity in AI access between genders and check whether growth in overall adoption is translating into a more equitable outcome.

Summary Takeaway

Across all eight business questions, one theme recurs: **AI adoption in education has grown enormously (roughly twelvefold in 12 years) but almost entirely uniformly** — across countries, regardless of infrastructure, and regardless of formal policy status. The equity gaps present at the start of the dataset (urban-rural, gender) have not closed as adoption has scaled; they have simply persisted alongside it. This is the central, decision-relevant insight for policymakers and edtech stakeholders reviewing this analysis. -e

Dashboard Design

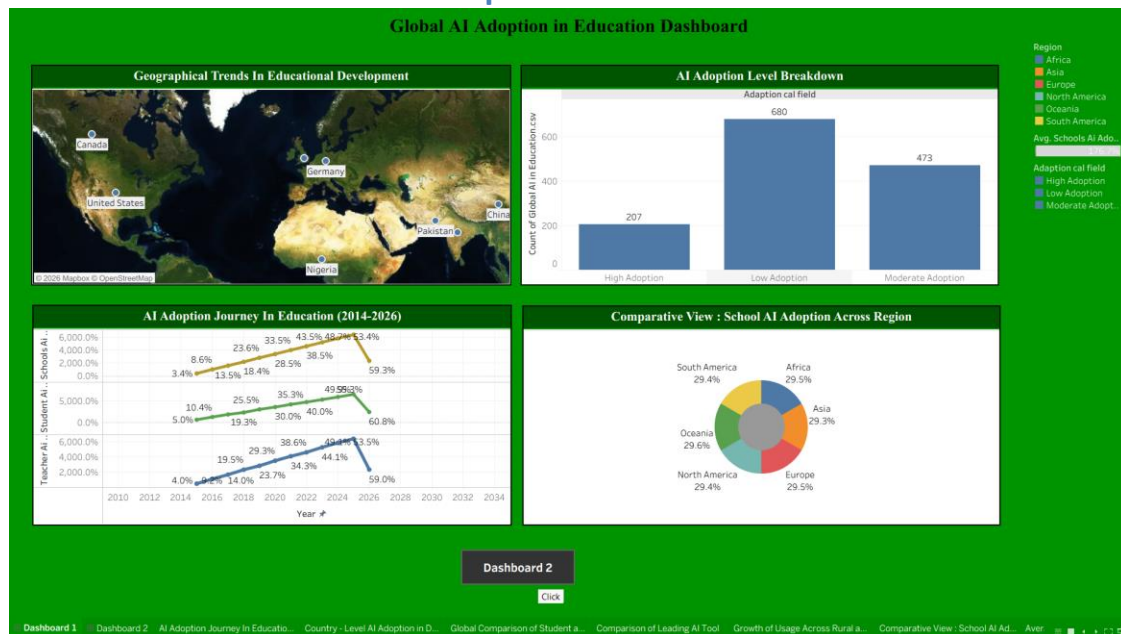
Global AI Adoption in Education

Field	Detail
Student Name	Shubham Vitthal Patil
Project Name	Global AI Adoption in Education: A Comprehensive Global Performance Intelligence Report
Team ID	Individual
Tool	Tableau Public

This dashboard design document presents the interactive Tableau visualizations built for the **Global AI Adoption in Education** project. The workbook consolidates 8 supporting worksheets into interactive dashboard views analysing 1,360 monthly records across 10 countries, 6 regions, and 16 attributes — covering adoption growth, regional patterns, tool preferences, and the urban-rural usage divide. Dashboards are interactive, filterable by Country/Region/Year/Government AI Policy, and cross-navigable via in-dashboard buttons.

🔗 **Published Dashboard (Tableau Public):** [Open Dashboard 1 on Tableau Public](#)

Dashboard 1 — “Global AI Adoption in Education Dashboard”



Dashboard 1

Chart 1 – Geographical Trends in Educational Development (Map)

- **Chart Type:** Satellite map with country markers.
- **Values Shown:** Marks the 10 countries in scope — Canada, United States, United Kingdom, Germany, China, Pakistan, India, Nigeria, Brazil, and Australia.
- **Purpose:** To give viewers an immediate geographic sense of which countries are covered before diving into the metrics.

Chart 2 – AI Adoption Level Breakdown (Bar Chart)

- **Chart Type:** Bar chart counting records by a calculated Adoption Tier field (Low 0–25% / Mid 26–45% / High 46–65%, based on `student_ai_usage_pct`).
- **Observation:** Low Adoption accounts for the most records (680), followed by Moderate Adoption (473), and High Adoption (207) — the majority of country-months in the dataset still fall in the lower adoption bands, since usage only crosses into “High” territory in the later years of the 2015–2026 span.
- **Purpose:** To classify the dataset into adoption tiers, showing that most of the observed history sits below the “High” tier even as the trend climbs sharply toward it by 2026.

Chart 3 – AI Adoption Journey in Education, 2014–2026 (Line Chart)

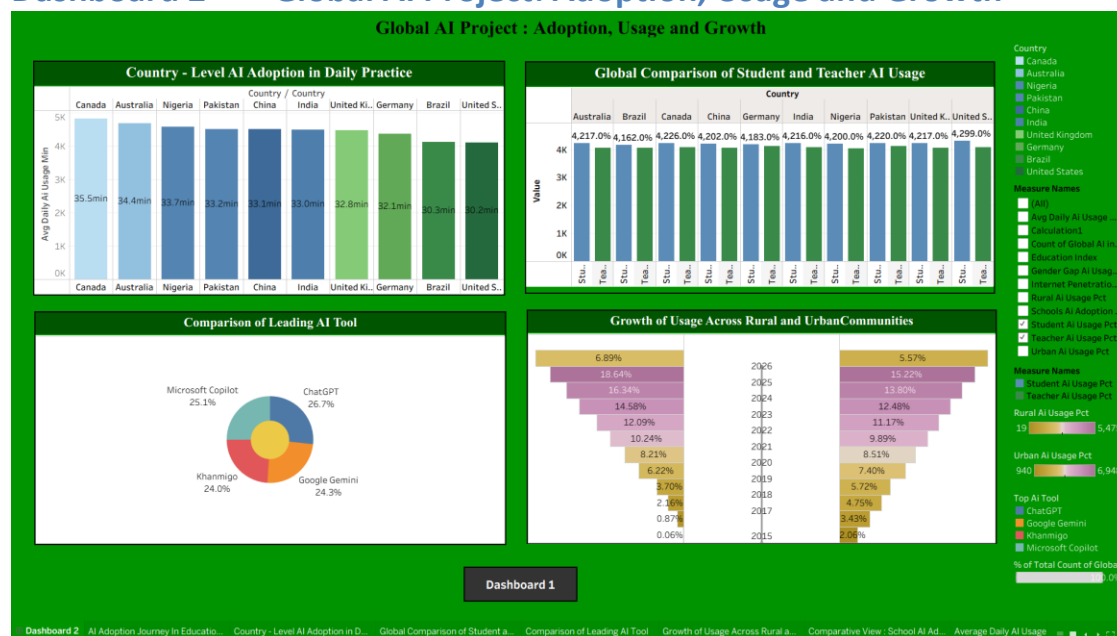
- **Chart Type:** Multi-line time series showing Student, Teacher, and School AI adoption % by year.
- **Observation:** All three lines climb in near lock-step from roughly 4.98% (student, 2015) to 60.75% by 2026 — a twelvefold increase — with student usage consistently leading teacher (59.00%) and school-level (59.25%) adoption by a small, stable margin throughout.

- **Purpose:** To visualize the multi-year growth trajectory of AI adoption across the three key stakeholder groups at once.

Chart 4 – Comparative View: School AI Adoption Across Region (Donut Chart)

- **Chart Type:** Donut chart showing average school AI adoption % by region.
- **Observation:** Adoption is nearly even across all six regions (~29.3–29.6% each). Because Asia’s total is built from 3 countries (China, India, Pakistan) versus 1–2 countries per other region, this chart is best read as a country-weighted average rather than a per-country comparison.
- **Purpose:** To show that regional averages are nearly uniform globally — the real variation in this dataset happens at the country level, not the region level.

Dashboard 2 — “Global AI Project: Adoption, Usage and Growth”



Dashboard 2

Chart 5 – Country-Level AI Adoption in Daily Practice (Bar Chart)

- **Chart Type:** Bar chart of average daily AI usage minutes by country.
- **Observation:** Canada leads at 35.5 min/day, followed by Australia (34.4 min) and Nigeria (33.7 min). The United States is lowest at 30.2 min, just behind Brazil (30.3 min).
- **Purpose:** To compare engagement *intensity* (time spent) rather than just adoption breadth across countries.

Chart 6 – Global Comparison of Student and Teacher AI Usage (Bar Chart)

- **Chart Type:** Grouped bar chart of summed Student vs. Teacher AI usage % by country.

- **Observation:** Totals are close across all 10 countries, with the United States posting the single highest summed value and Brazil the lowest — student usage totals edge out teacher usage totals in every country, but only slightly, consistent with the near-uniform country-level averages seen in the Country vs. Student AI Usage chart (30.60%–31.61%).
- **Purpose:** To benchmark cumulative student vs. teacher AI engagement side-by-side across countries.

Chart 7 – Comparison of Leading AI Tool (Donut Chart)

- **Chart Type:** Donut chart showing the share of records where each tool is the top AI tool used.
- **Observation:** ChatGPT leads with 26.7% of records, followed by Microsoft Copilot (25.1%), Google Gemini (24.3%), and Khanmigo (24.0%) — a near-even four-way split, with Asia and Africa showing the broadest spread and ChatGPT edging ahead in the more recent months of the dataset.
- **Purpose:** To show that the global AI-in-education tool market is fragmented rather than monopolized by one platform.

Chart 8 – Growth of Usage Across Rural and Urban Communities (Diverging Bar Chart)

- **Chart Type:** Diverging bar chart (population-pyramid style) comparing Rural (left) vs. Urban (right) AI usage share by year, 2015–2026.
- **Observation:** The Urban-Rural Usage Gap widened sharply from 7.68 points in 2015 to roughly 12–13 points by 2018, and has stayed essentially flat since (12.98 points in 2026) — meaning the gap has plateaued rather than closed even as overall usage keeps climbing.
- **Purpose:** To track how the rural-urban usage split has shifted year over year, and confirm the equity gap is persistent rather than narrowing.

Dashboard Navigation

Dashboard 1 and Dashboard 2 are cross-linked via in-dashboard navigation buttons (“Dashboard 2” / “Dashboard 1”), allowing viewers to move between the adoption-trend view and the usage/tool-comparison view without leaving the workbook. Both dashboards share global filters for Country, Region, Year, and Government AI Policy status. -e

Story

Global AI Adoption in Education

Field	Detail
Student Name	Shubham Vitthal Patil

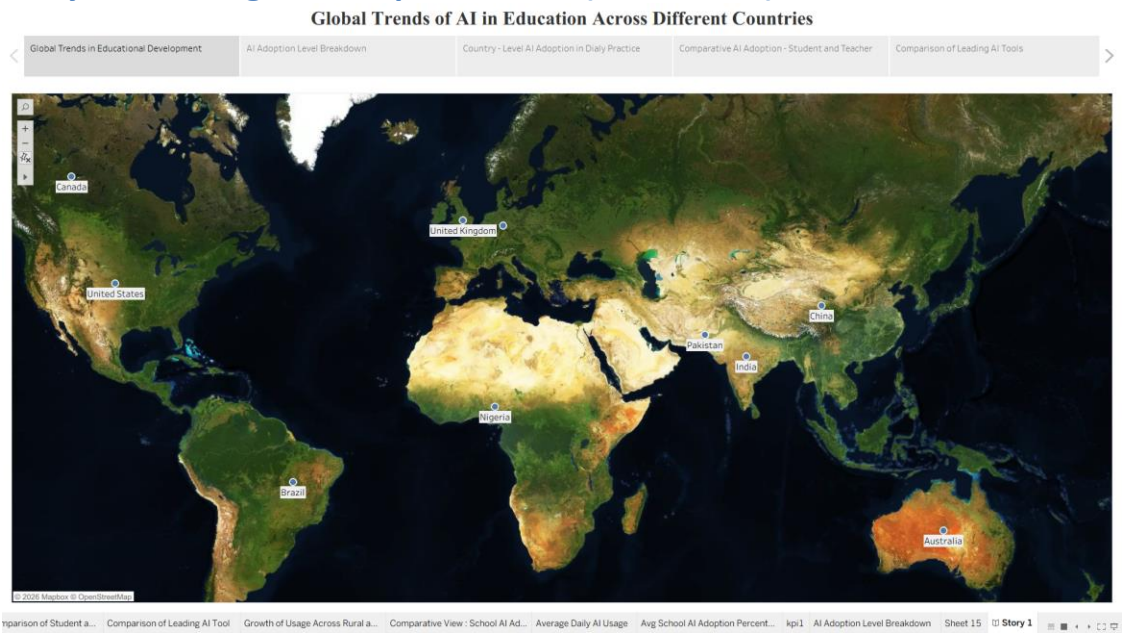
Field	Detail
Project Name	Global AI Adoption in Education: A Comprehensive Global Performance Intelligence Report
Team ID	Individual
Tool	Tableau Public

By using stories in Tableau, complex data can be communicated in a way that is both interactive and engaging, making it easier for the audience to follow along and understand the insights. It’s a tool for **data storytelling**, allowing insights to be presented in a cohesive, logical progression.

In Tableau, **Story** is a feature that allows a sequence of dashboards, visualizations, and text to present data insights in a narrative-driven way — guiding the audience through the key trends and outcomes of the analysis.

 **Published Story (Tableau Public):** [Open “Decoding the Adoption Curve” Story on Tableau Public](#)

Story: “Decoding the Adoption Curve (2015–2026)”



Story Overview

The story unfolds across five narrative points, moving from the broad global growth curve to a specific, surprising closing conclusion about policy and equity:

Point 1 — AI Adoption Trend by Year

- **Visualization:** Full time-series view of student, teacher, and school AI adoption from 2015 to 2026.
- **Narrative Role:** Opens the story with the headline growth number — student usage climbs from 4.98% in 2015 to 60.75% by 2026, roughly a twelvefold increase, with teacher (59.00%) and school-level (59.25%) adoption converging closely behind by the final year.

Point 2 — Country vs Student AI Usage (%)

- **Visualization:** 10-bar chart ranking every country by average student AI usage.
- **Narrative Role:** Shifts from “how much” to “who” — led by the United States (31.61%) and Canada (31.07%), tapering down to Brazil (30.60%) at the lower end, revealing a visibly tight cluster rather than a wide spread between countries.

Point 3 — Internet Penetration & Education Index vs AI Adoption

- **Visualization:** Scatter plot with trend line.
- **Narrative Role:** Tests and disproves an intuitive assumption — Internet Penetration and Education Index correlate strongly with each other ($r \approx 0.98$), but neither correlates meaningfully with Student, Teacher, or School AI Usage ($r \approx 0.00$ in all cases), undermining the idea that better-connected or higher-scoring education systems necessarily adopt AI faster.

Point 4 — Government Policy vs Curriculum Integration

- **Visualization:** Stacked bar chart comparing Policy Status (Active / Draft / None) against curriculum integration rate.
- **Narrative Role:** Drives home a counter-intuitive finding — curriculum-integration rates are nearly identical whether a country has an Active policy (52.3%), a Draft policy (47.2%), or no recorded policy at all (51.9%), suggesting formal policy alone is not moving the needle on classroom practice.

Point 5 — Equity Gaps Over Time (Closing Insight)

- **Visualization:** Combined trend view of the Urban-Rural Usage Gap and the Gender Gap in AI Usage, 2015–2026.
- **Narrative Role:** The story’s key “punchline” — the urban-rural usage gap widened from 7.68 points in 2015 to roughly 12–13 points by 2018 and has stayed essentially flat since (12.98 points in 2026); the gender gap has similarly held steady between 5.05 and 5.91 points across the entire period, with no clear narrowing trend.

Overall Narrative Takeaway

Overall AI adoption in education has grown enormously over twelve years, but the equity gaps that existed early on — urban vs. rural, and gender — have not closed alongside it. Growth has been **additive rather than equalizing**, implying that policy and investment specifically targeted at rural and gender-gap closure — not just general AI rollout — will be needed to bring these gaps down. -e

